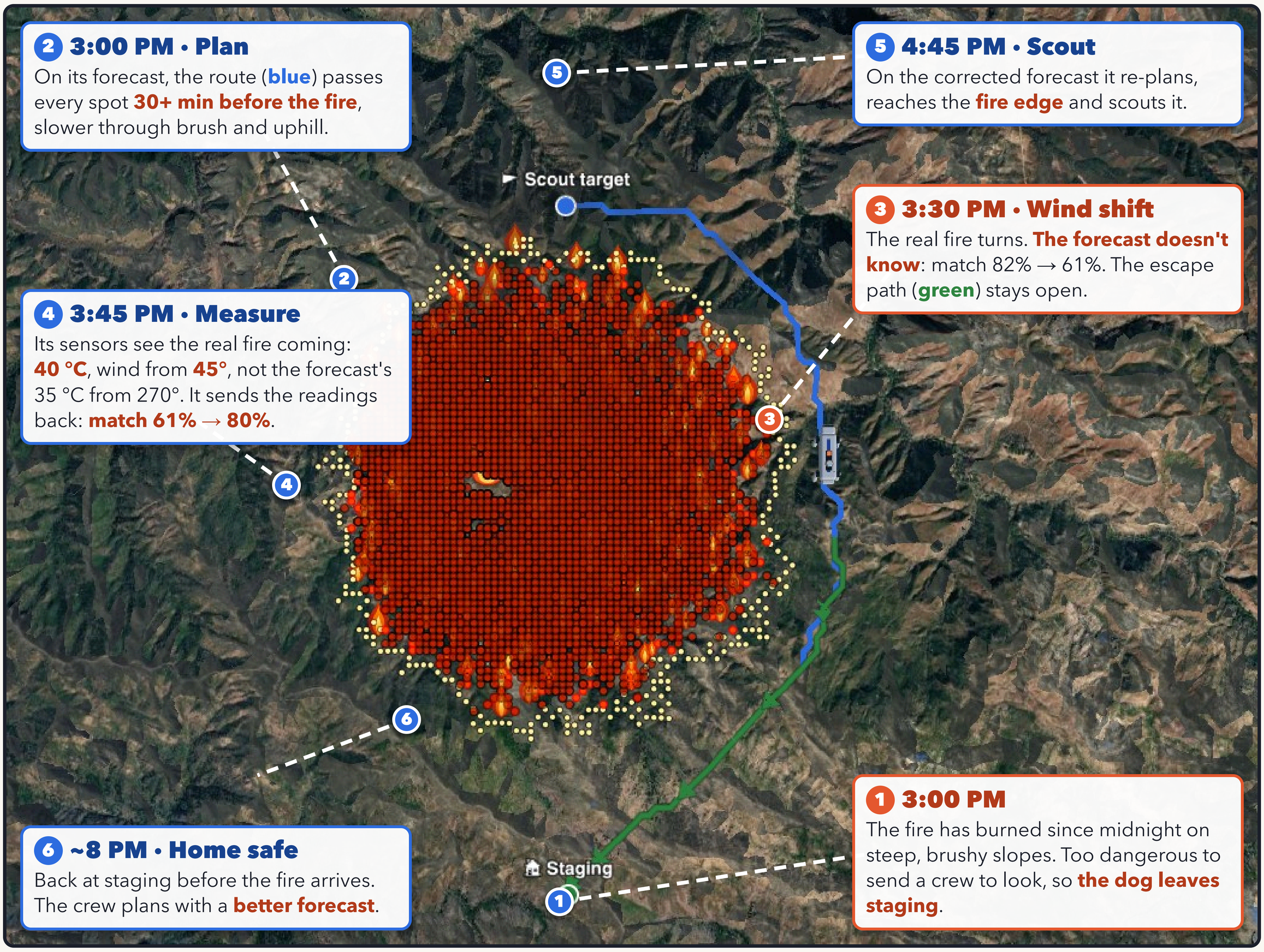
Fire-Aware Robot Dogs

Machine-Learning Wildfire Forecasts for Safe Navigation and Scouting

Shawn Zeng · Monta Vista High School

One afternoon at a Big Sur fire

A simulated mission on real terrain. Follow the dog ① → ⑥.



📡 **Stop ④: what the dog measured**
vs. what the forecast assumed

Wind	Temperature	Humidity	Fire edge	Forecast match
45° 25 km/h	**40 °C**	**5 %**	**20 m closer**	**61% → 80%**
forecast 270° 15	forecast 35 °C	forecast 5 %	than forecast	next 4 h of spread

How the dog decides at each stop, and how we tested it: **right panel →**

Know where the fire *will* be – not just where it is.

- 👨‍🚒 **Safer crews**
  Fewer firefighters sent into danger to look.
- 🛡️ **Robot stays safe**
  Routes and exits keep it from getting trapped.
- 📍 **Smarter sensing**
  Measure where it changes decisions.